

Lab 2

PingFederate Connection

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Environment: Ping Identity DevOps Docker Lab - PingDirectory · PingFederate · PingAccess

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1. Objective

This lab connects PingFederate to the CyberCorp PingDirectory environment built in Lab 1, establishing the authentication path for a fictitious application, the Cascade Internal Portal. It covers registering PingDirectory as an LDAP data store inside PingFederate, creating an LDAP-backed credential validator, building an HTML Form IdP Adapter that challenges the user for a username and password, defining an Authentication Policy Contract that carries the authenticated user's attributes forward, and assembling an Authentication Policy that chains the adapter to that contract into a single, working login flow.

Every component created here reuses the identity foundation from Lab 1: the svc-pingfederate service account and the ou=People branch of dc=cybercorp,dc=local.

2. Lab Environment Verification

Before configuring anything, the running PingFederate container and the admin console were confirmed reachable.

```
devanshu@ping-lab:~$ docker ps | grep pingfederate
85ade2b05e0a pingidentity/pingfederate:latest "/bootstrap.sh star..." 5 days ago
Up 12 hours (healthy) 0.0.0.0:9031->9031/tcp, [::]:9031->9031/tcp, 0.0.0.0:999
9->9999/tcp, [::]:9999->9999/tcp pingfederate
devanshu@ping-lab:~$ |
```

Figure 1. Confirming the pingfederate container is up and healthy (docker ps | grep pingfederate).

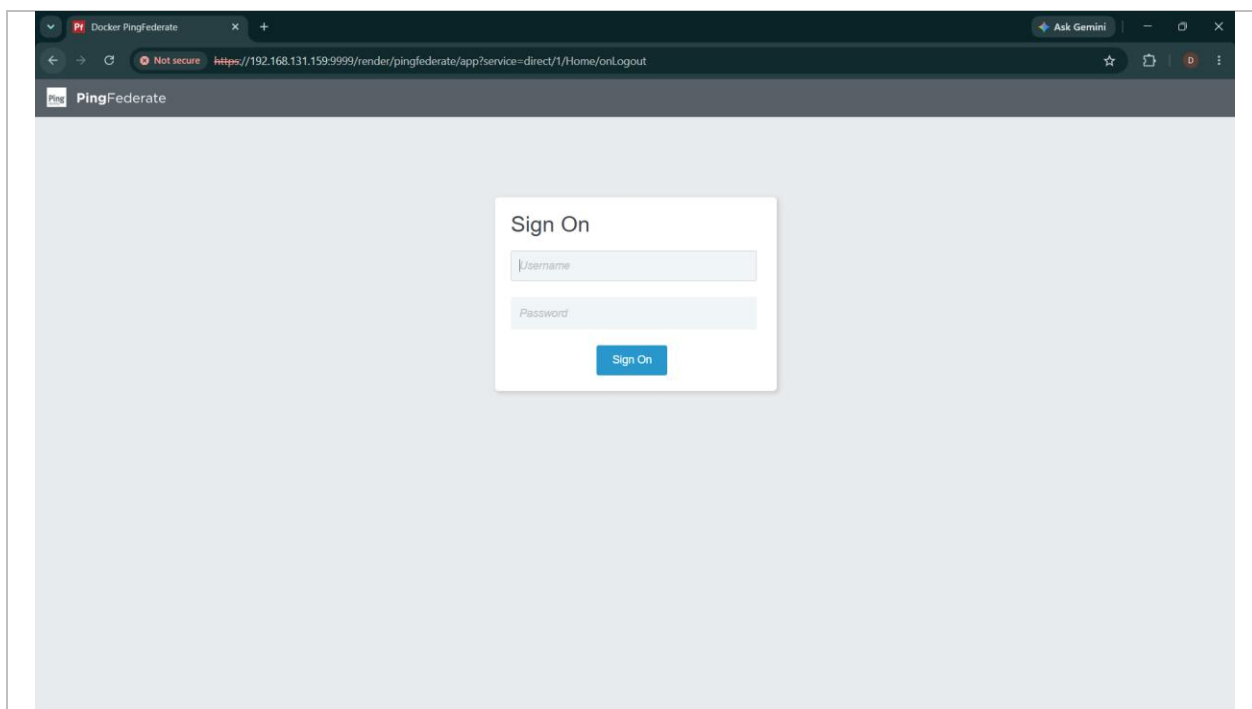


Figure 2. The PingFederate Sign On page, reached over HTTPS on port 9999.

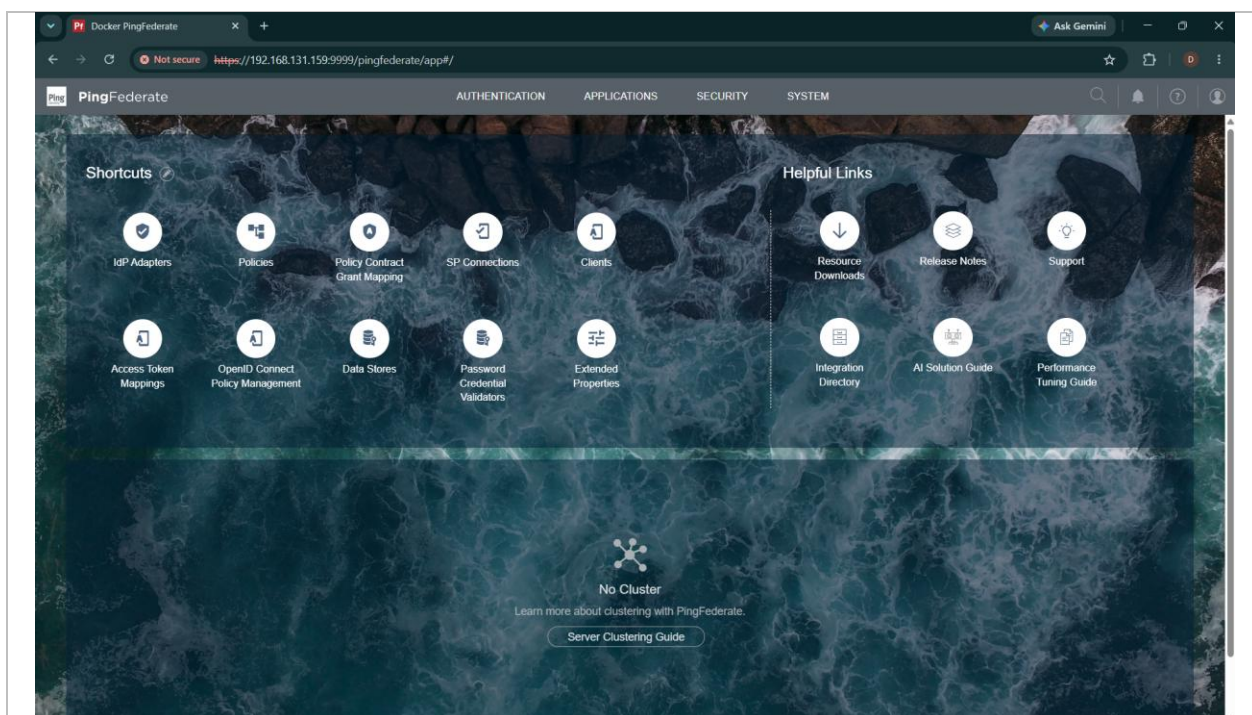


Figure 3. The PingFederate administrative console home page, showing shortcuts to IdP Adapters, Policies, Data Stores, and Password Credential Validators.

Note: The PingFederate admin console uses the default demo credentials shipped with the Docker image (Administrator / 2FederationM0re). As with the svc-pingfederate directory password from Lab 1, treat these as throwaway lab credentials and rotate or redact anything similar before this environment is anything other than a local, disposable lab.

3. Connecting PingFederate to the CyberCorp Directory

Five components were created, each building on the last: a data store pointing at PingDirectory, a credential validator that uses it to check passwords, an IdP Adapter that presents the login form, a policy contract that defines the attributes a relying application receives, and an authentication policy that ties the adapter and the contract together.

Component	Name
LDAP Data Store	CyberCorp-PD
Password Credential Validator	CyberCorp-LDAP-Validator
IdP Adapter	Cascade-Portal-Login
Authentication Policy Contract	CyberCorp-User-Authentication
Authentication Policy	CyberCorp-Cascade-Authentication

3.1 Creating the PingDirectory LDAP Data Store

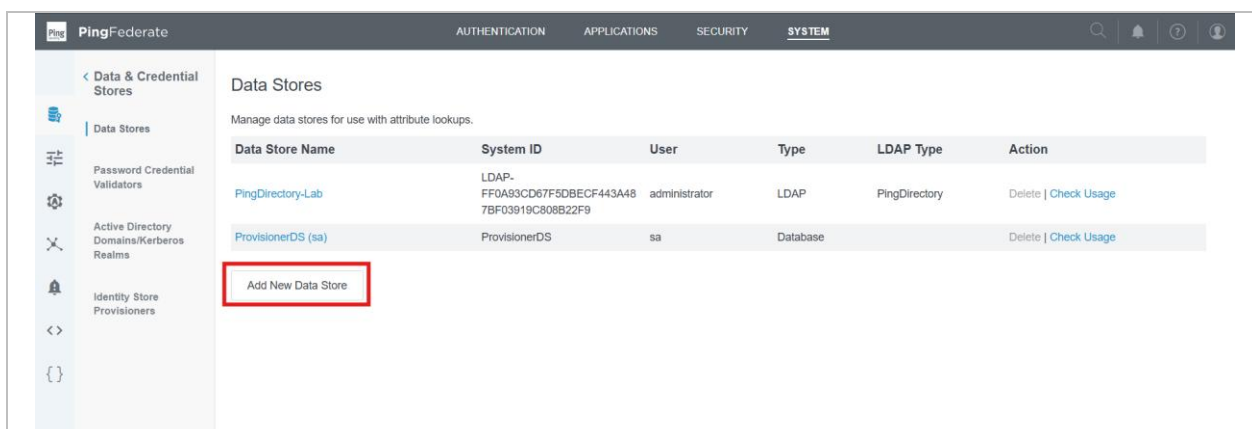


Figure 4. Data Stores list, showing the pre-existing PingDirectory-Lab and ProvisionerDS stores, before adding a new one.

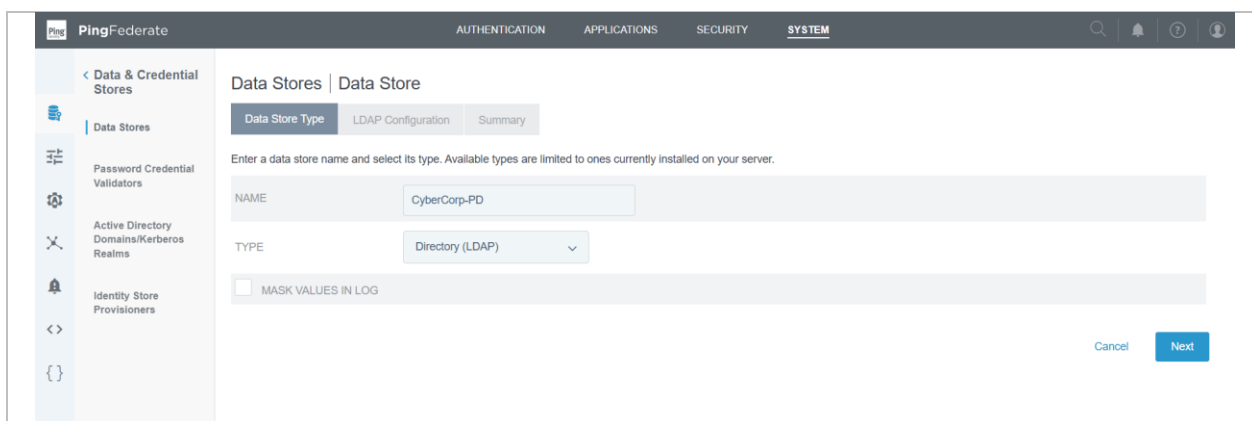


Figure 5. Naming the new data store CyberCorp-PD and setting its type to Directory (LDAP).

Ping

PingFederate

AUTHENTICATIONAPPLICATIONSSECURITYSYSTEM

< Data & Credential StoresData StoresPassword Credential ValidatorsActive Directory Domains/Kerberos RealmsIdentity Store Provisioners<>{ }

Data Stores | Data Store

Data Store TypeLDAP ConfigurationSummary

Complete the configuration necessary to connect to your data store. For regional deployments, specify region-specific hostnames on separate rows and the corresponding node tags for each region. To tag the nodes that use a particular hostname, use the `node.tags` property in the `run.properties` file. Failover can be configured within a single region by listing multiple space-separated hostnames in the same hostname row. Failover is not supported across different rows.

Hostname(s)	Tags	Action
localhost:1389		Edit Undelete
pingdirectory:1389	old	Edit Delete Default
		Add

CONNECTION SECURITY

☒ None☐ LDAPS☐ StartTLS

☐ USE DNS SRV RECORD

LDAP TYPE

PingDirectory

AUTHENTICATION METHOD

☐ None (Anonymous)☒ Simple☐ Client TLS Certificate

CREDENTIAL STORAGE

☒ Internally Managed☐ Secret Manager

USER DN

uid=svc-pingfederate,ou

PASSWORD

pingdirectory:1389

Test Connection

Connectivity test was successful.

Manage Secret Managers

Manage SSL Client Keys & Certificates

Advanced

Cancel

Previous

Next

Save

Figure 6. LDAP Configuration: hostname pingdirectory:1389, Connection Security None, LDAP Type PingDirectory, Simple authentication bound as uid=svc-pingfederate, and a successful Test Connection.

PingFederate AUTHENTICATION APPLICATIONS SECURITY **SYSTEM**

Data Stores | Data Store

Data Store Type LDAP Configuration Summary

Click a heading link to edit a configuration setting.

Data Store

Data Store Type

Type of Data Store LDAP

LDAP Configuration

Data Store Name CyberCorp-PD

Hostname(s) Hostname(s): pingdirectory:1389
Tags: old
Default

LDAP Type PingDirectory

Connection Security None

Authentication Method Simple

Username uid=svc-pingfederate,ou=ServiceAccounts,dc=cybercorp,dc=local

Mask Values in Log false

Retry Failed Operations false

Test Connection on Borrow false

Test Connection on Return false

Create New Connections if Necessary true

Verify LDAPS Hostname true

Minimum Connections 10

Maximum Connections 100

Maximum Wait (ms) -1

Time Between Eviction (ms) 60000

Read Timeout (ms) 3000

Connection Timeout (ms) 3000

DNS TTL (ms) 60000

LDAP DNS SRV Record prefix _ldap_tcp

LDAPS DNS SRV Record prefix _ldaps_tcp

Cancel Previous **Save**

Figure 7. Data store summary confirming CyberCorp-PD before saving.

What this does

CyberCorp-PD registers the CyberCorp PingDirectory instance as an LDAP data store inside PingFederate. This is the connection every later component reuse whenever it needs to reach into PingDirectory, whether to check a password or to look up additional user attributes.

It points at `pingdirectory:1389` using Simple authentication, bound as `uid=svc-pingfederate,ou=ServiceAccounts,dc=cybercorp,dc=local` - the same read-only service account created and secured with an ACL in Lab 1. Connectivity was confirmed with the built-in Test Connection tool before saving, so any later failures can be isolated to configuration rather than network or credential problems.

3.2 Creating the LDAP Password Credential Validator

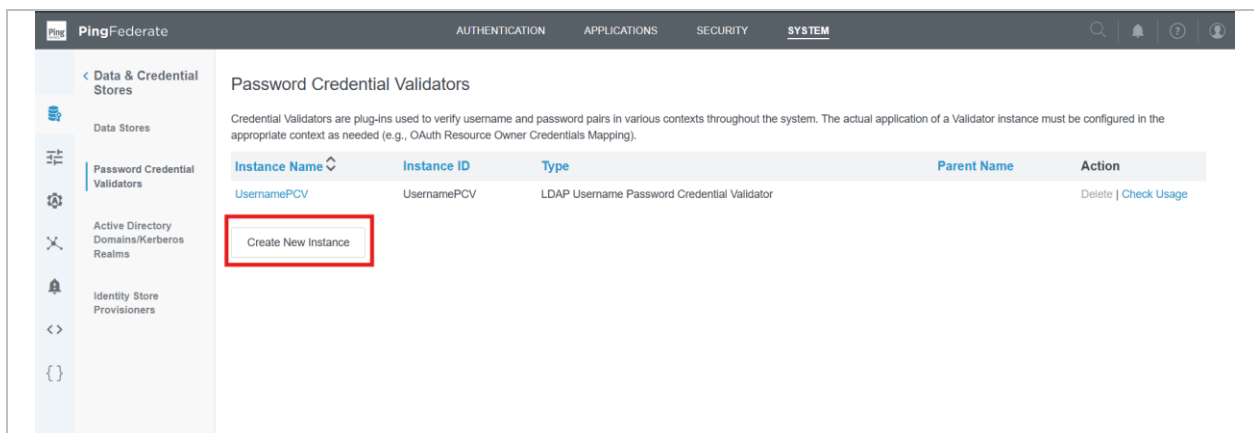


Figure 8. Password Credential Validators list, showing the existing UsernamePCV instance, before creating a new one.

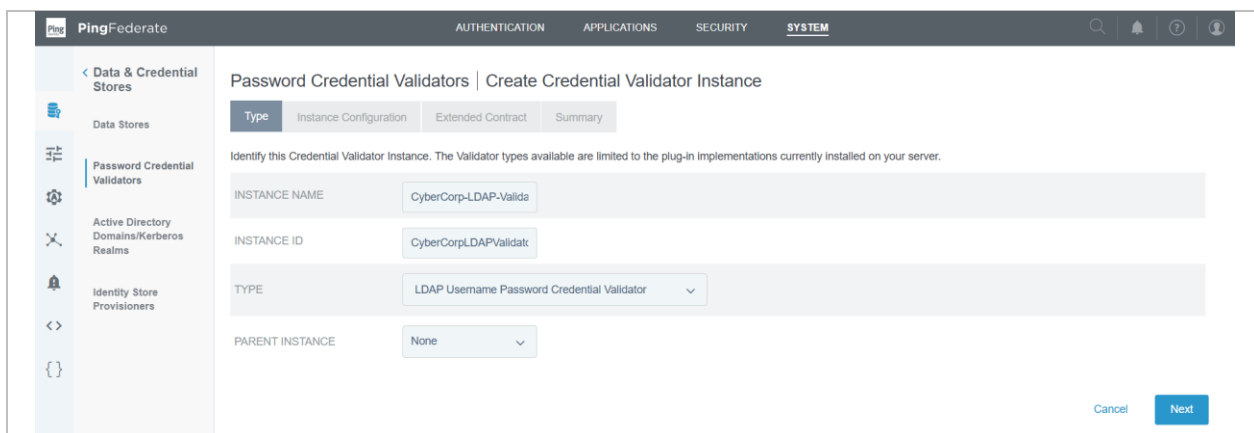


Figure 9. Naming the new validator CyberCorp-LDAP-Validator, type LDAP Username Password Credential Validator.

PingFederate AUTHENTICATION APPLICATIONS SECURITY **SYSTEM**

Password Credential Validators | Create Credential Validator Instance

Type **Instance Configuration** Extended Contract Summary

Complete the configuration necessary for this Password Credential Validator to check username/password pairs. This configuration was designed into, and is specific to, the selected Credential Validator plug-in.

This password credential validator provides a means of verifying credentials stored in a directory server via the LDAP protocol. Additional user attributes from the directory can also be returned by this PCV by adding the desired attribute names to the Extended Contract.

Authentication Error Overrides ⓘ

Match Expression ⓘ	Error	Message Properties Key ⓘ	Action
Add a new row to 'Authentication Error Overrides'			

Field Name	Field Value	Description
LDAP DATASTORE	CyberCorp-PD	Select the LDAP Datastore.
SEARCH BASE	ou=People,dc=cybercorp,dc=local	The location in the directory from which the LDAP search begins.
SEARCH FILTER	((uid=\${username})(mail=\${username}))	You may use \${username} as part of the query. Example (for Active Directory): sAMAccountName=\${username}.
SCOPE OF SEARCH	☐ One Level ☒ Subtree	
CASE-SENSITIVE MATCHING	☒	Allows case-sensitive expression and LDAP error matching.

[Manage Data Stores](#) [Show Advanced Fields](#)

[Cancel](#) [Previous](#) [Next](#)

Figure 10. Instance configuration: LDAP Datastore CyberCorp-PD, Search Base ou=People,dc=cybercorp,dc=local, Search Filter matching uid or mail against the submitted username, Subtree scope.

PingFederate AUTHENTICATION APPLICATIONS SECURITY **SYSTEM**

Password Credential Validators | Create Credential Validator Instance

Type **Instance Configuration** **Extended Contract** Summary

You can extend the attribute contract of this Password Credential Validator instance.

Core Contract

DN

givenName

mail

username

Extend the Contract

	Action
cn	Edit Delete
departmentNumber	Edit Delete
title	Edit Delete
uid	Edit Delete
	Add

[Cancel](#) [Previous](#) [Next](#)

Figure 11. Extended Contract: the default DN, givenName, mail, and username, plus cn, departmentNumber, title, and uid added from the PingDirectory schema.

PingFederate AUTHENTICATION APPLICATIONS SECURITY **SYSTEM**

Password Credential Validators | Create Credential Validator Instance

Type Instance Configuration Extended Contract **Summary**

Password Credential Validator configuration summary.

Create Credential Validator Instance

Type

Instance Name	CyberCorp-LDAP-Validator
Instance ID	CyberCorpLDAPValidator
Type	LDAP Username Password Credential Validator
Class Name	org.sourceid.saml20.domain.LDAPUsernamePasswordCredentialValidator
Parent Instance Name	None

Instance Configuration

LDAP Datastore	CyberCorp-PD
Search Base	ou=People,dc=cybercorp,dc=local
Search Filter	((uid=\${username}))(mail=\${username}))
Scope of Search	Subtree
Case-Sensitive Matching	true
Display Name Attribute	displayName
Mail Attribute	mail
SMS Attribute	
PingID Username Attribute	
Mail Search Filter	
Username Attribute	
Trim Username Spaces For Search	true
Mail Verified Attribute	
Account Disabled Attribute	
Enable PingDirectory Detailed Password Policy Requirement Messaging	true
Expect Password Expired Control	false

Extended Contract

Attribute	DN
Attribute	givenName
Attribute	mail
Attribute	username
Attribute	cn
Attribute	departmentNumber
Attribute	title
Attribute	uid

Cancel Previous **Save**

Figure 12. Credential validator summary confirming CyberCorp-LDAP-Validator before saving.

What this does

CyberCorp-LDAP-Validator is the component PingFederate actually calls to check a submitted username and password. It's built on the CyberCorp-PD data store from Step 3.1, searching `ou=People,dc=cybercorp,dc=local` for an entry whose uid or mail matches the submitted username, then validating the supplied password against that entry over an LDAP simple bind.

We also extended the validator's contract with cn, departmentNumber, title, and uid on top of the default DN, givenName, mail, and username. That way, once a user authenticates, those PingDirectory attributes are immediately available to the rest of the flow without requiring a second directory lookup.

3.3 Creating the IdP Adapter

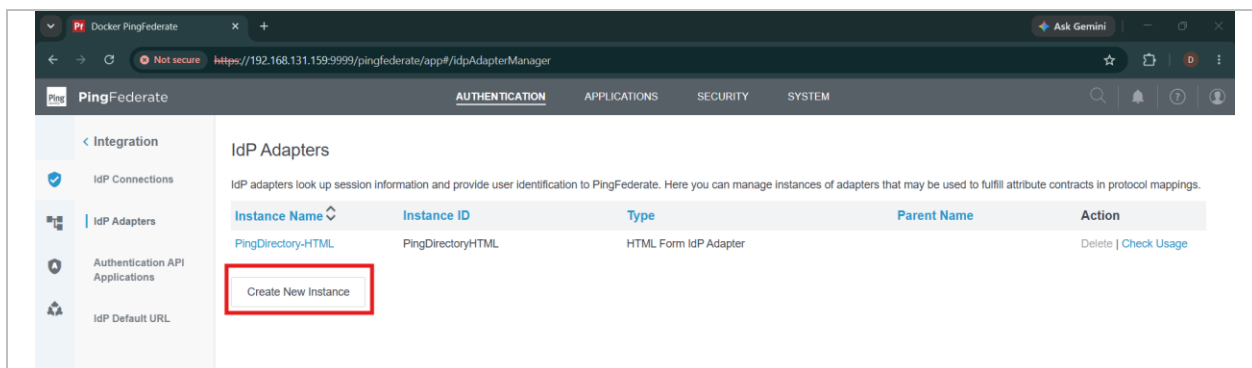


Figure 13. IdP Adapters list, showing the existing PingDirectory-HTML adapter, before creating a new one.

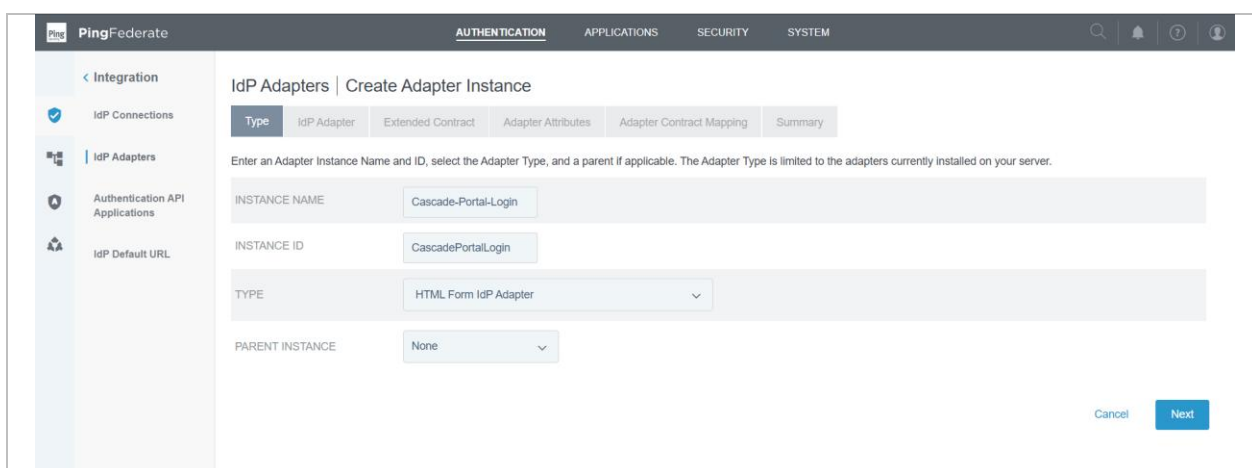


Figure 14. Naming the new adapter Cascade-Portal-Login, type HTML Form IdP Adapter.

AUTHENTICATION
APPLICATIONS
SECURITY
SYSTEM

Integration

IdP Connections

IdP Adapters

Authentication API Applications

IdP Default URL

IdP Adapters | Create Adapter Instance

Type
IdP Adapter
Extended Contract
Adapter Attributes
Adapter Contract Mapping
Summary

Complete the configuration necessary to look up user security contexts in your environment. This configuration was designed into the adapter for use at your site.

Credential Validators ⓘ

Password Credential Validator Instance

CyberCorp-LDAP-Validator

Update | Cancel

Add a new row to 'Credential Validators'

Field Name	Field Value	Description
CHALLENGE RETRIES	3	Number of failed user authentications after which the PingFederate account locking service blocks future attempts.
SESSION STATE	☐ Globally ☐ Per Adapter ☒ None	Determines how state is maintained within one adapter or between different adapter instances. To take advantage of additional features, it is recommended to use a PingFederate Authentication Session rather than this adapter's internal Session State capability.
SESSION TIMEOUT	60	Session Idle Timeout (in minutes). If left blank the timeout will be the Session Max Timeout. Ignored if 'None' is selected for Session State.
SESSION MAX TIMEOUT	480	Session Max Timeout (in minutes). Leave blank for indefinite sessions. Ignored if 'None' is selected for Session State.
ALLOW PASSWORD CHANGES	☐	Allows users to change their password using this adapter.
PASSWORD MANAGEMENT SYSTEM		A fully-qualified URL to your password management system where users can change their password. If left blank, password changes are handled by this adapter.
ENABLE 'REMEMBER MY USERNAME'	☐	Allows users to store their username as a cookie when authenticating with this adapter. Once stored, the username is pre-populated in the login form's username field on subsequent transactions.
ENABLE 'THIS IS MY DEVICE'	☐	Allows users to indicate whether their device is shared or private. In this mode, PingFederate Authentication Sessions will not be stored unless the user indicates the device is private. This adapter's internal session tracking (if enabled) will not be affected by the user's selection.
CHANGE PASSWORD POLICY CONTRACT	Select	The policy contract to use for change password. Selecting a policy contract will enable usage of Authentication Policy during password change.
CHANGE PASSWORD NOTIFICATION	☐	Send users a notification upon a password change. This feature relies on the underlying PCV returning 'mail' and 'givenName' attributes containing the user's first name and e-mail address. Additionally, a notification publisher must be configured.
SHOW PASSWORD EXPIRING WARNING	☐	Show a warning message to the user on login about an approaching password expiration.
PASSWORD RESET TYPE	☐ Authentication Policy ☐ Email One-Time Link ☐ Email One-Time Password ☐ PingID ☐ Text Message ☒ None	Select the method to use for self-service password reset. Depending on the selected method, additional settings are required to complete the configuration. It is recommended that the method used is not already part of a multi-factor authentication policy that includes a password challenge, as that would indirectly reduce that authentication policy to a single factor. For example, if users normally authenticate with a password challenge and then PingID, the self-service password reset method should not be PingID.
PASSWORD RESET POLICY CONTRACT	Select	The policy contract to use for password reset. This is used for the password reset type 'Authentication Policy'.
REVOKE SESSIONS AFTER PASSWORD CHANGE OR RESET	☐	Revokes a user's authentication sessions after a password change or reset is completed by this adapter. This option relies on selecting a unique user key attribute for this adapter.
ACCOUNT UNLOCK	☐	Allows users with a locked account to unlock it using the self-service password reset type.
LOCAL IDENTITY PROFILE	-- Select One --	Optionally associate this instance with a Local Identity Profile.
NOTIFICATION PUBLISHER	-- Select One --	Optionally associate this instance with a notification delivery mechanism.
ENABLE USERNAME RECOVERY	☐	Allow users to get their username from an email.

Manage Password Credential Validators
Manage SMS Provider Settings
Manage Local Identity Profiles
Manage Notification Publishers

Manage CAPTCHA and Risk Providers
Manage Policy Contracts
Show Advanced Fields

Cancel
Previous
Next

Figure 15. IdP Adapter configuration: Credential Validator set to CyberCorp-LDAP-Validator, Session State None, Session Timeout 60 minutes, Session Max Timeout 480 minutes, all other fields left at their defaults.

The screenshot shows the 'Create Adapter Instance' page in the PingFederate console, specifically the 'Extended Contract' tab. The left sidebar shows the navigation menu with 'IdP Adapters' selected. The main content area has a tabbed interface with 'Type', 'IdP Adapter', 'Extended Contract' (active), 'Adapter Attributes', 'Adapter Contract Mapping', and 'Summary'. Below the tabs, a text block explains that this adapter type supports creating an extended adapter contract after initial deployment. The 'Core Contract' section shows 'policy.action' and 'username' as default attributes. The 'Extend the Contract' section has an empty input field and an 'Add' button. At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

Figure 16. Extended Contract left at its defaults: *policy.action* and *username*, with no additional attributes added at this stage.

The screenshot shows the 'Adapter Attributes' tab of the 'Create Adapter Instance' page. The 'UNIQUE USER KEY ATTRIBUTE' dropdown is set to 'username'. Below this is a table for configuring attributes:

Attribute	Pseudonym	Mask Log Values
policy.action	☐	☐
username	☒	☐

At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

Figure 17. Adapter Attributes: *username* set as both the Unique User Key Attribute and the pseudonym value.

The screenshot shows the 'Adapter Contract Mapping' tab of the 'Create Adapter Instance' page. The 'Configure Adapter Contract' button is highlighted with a red rectangle. Below the tabs, a text block explains that an Adapter Contract may be used to fulfill the Attribute Contract passed to SP partners. At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

Figure 18. Adapter Contract Mapping: choosing to configure the adapter contract.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

Integration

- IdP Connections
- IdP Adapters
- Authentication API Applications
- IdP Default URL

IdP Adapters | Create Adapter Instance | Adapter Contract Mapping

Attribute Sources & User Lookup | Adapter Contract Fulfillment | Issuance Criteria | Summary

You can choose to fulfill the Adapter Contract with the adapter's default values, or you can use these values plus additional attributes retrieved from local data stores.

Description	Type	Action
Add Attribute Source		

Cancel Next

Figure 19. Attribute Sources & User Lookup left empty — no additional data store is queried at the adapter level.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

Integration

- IdP Connections
- IdP Adapters
- Authentication API Applications
- IdP Default URL

IdP Adapters | Create Adapter Instance | Adapter Contract Mapping

Attribute Sources & User Lookup | Adapter Contract Fulfillment | Issuance Criteria | Summary

Fulfill your Adapter Contract with values from the authentication adapter or with dynamic text values.

Contract	Source	Value	Actions
policy.action	Adapter		None available
username	Adapter		None available

Cancel Previous Next

Figure 20. Adapter Contract Fulfillment: both *policy.action* and *username* are fulfilled directly by the Adapter, with no external source.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

Integration

- IdP Connections
- IdP Adapters
- Authentication API Applications
- IdP Default URL

IdP Adapters | Create Adapter Instance | Adapter Contract Mapping

Attribute Sources & User Lookup | Adapter Contract Fulfillment | Issuance Criteria | Summary

PingFederate can evaluate various criteria to determine whether users are authorized to access SP resources. Use this optional screen to configure the criteria for use with this conditional authorization.

Source	Attribute Name	Condition	Value	Error Result	Action
- SELECT -	- SELECT -	- SELECT -			Add

Cancel Previous Next

Figure 21. Issuance Criteria left unconfigured — no conditional authorization rules are applied at this stage.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

< Integration

- IdP Connections
- IdP Adapters
- Authentication API Applications
- IdP Default URL

IdP Adapters | Create Adapter Instance | Adapter Contract Mapping

Attribute Sources & User Lookup | Adapter Contract Fulfillment | Issuance Criteria | **Summary**

Click a heading link to edit a configuration setting.

Adapter Contract Mapping

Attribute Sources & User Lookup

Data Sources	(None)
--------------	--------

Adapter Contract Fulfillment

policy.action	policy.action (Adapter)
username	username (Adapter)

Issuance Criteria

Criterion	(None)
-----------	--------

Cancel Previous **Done**

Figure 22. Adapter Contract Mapping summary before returning to the adapter wizard.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

< Integration

- IdP Connections
- IdP Adapters
- Authentication API Applications
- IdP Default URL

IdP Adapters | Create Adapter Instance

Type | IdP Adapter | Extended Contract | Adapter Attributes | Adapter Contract Mapping | **Summary**

IdP adapter instance summary information.

Create Adapter Instance

Type

Instance Name	Cascade-Portal-Login
Instance ID	CascadePortalLogin
Type	HTML Form IdP Adapter
Class Name	com.pingidentity.adapters.htmlform.idp.HtmlFormIdpAuthnAdapter
Parent Instance Name	None

IdP Adapter

Credential Validators	CyberCorp-LDAP-Validator
Challenge Retries	3
Session State	None
Session Timeout	60
Session Max Timeout	480
Allow Password Changes	false
Password Management System	
Enable 'Remember My Username'	false
Enable 'This is My Device'	false

Cancel Previous **Done**

Figure 23. IdP Adapter summary (part 1): Cascade-Portal-Login, class HtmlFormIdpAuthnAdapter, Credential Validator CyberCorp-LDAP-Validator, session settings.

PingFederate		AUTHENTICATION	APPLICATIONS	SECURITY	SYSTEM
< Integration					
IdP Connections	Change Password Policy Contract				
	Change Password Notification	false			
IdP Adapters	Show Password Expiring Warning	false			
	Password Reset Type	None			
Authentication API Applications	Password Reset Policy Contract				
	Revoke Sessions After Password Change Or Reset	false			
IdP Default URL	Account Unlock	false			
	Local Identity Profile	None Selected			
	Notification Publisher	None Selected			
	Enable Username Recovery	false			
	Login Template	html.form.login.template.html			
	Logout Path				
	Logout Redirect				
	Logout Template	idp.logout.success.page.template.html			
	Change Password Template	html.form.change.password.template.html			
	Change Password Message Template	html.form.message.template.html			
	Password Management System Message Template	html.form.message.template.html			
	Password Update Timeout	30			
	Change Password Email Template	message-template-end-user-password-change.html			
	Expiring Password Warning Template	html.form.password.expiring.notification.template.html			
	Threshold for Expiring Password Warning	7			
	Snooze Interval for Expiring Password Warning	24			

Figure 24. IdP Adapter summary (part 2): password-management and reset options, all left at their defaults (disabled).

PingFederate		AUTHENTICATION	APPLICATIONS	SECURITY	SYSTEM
< Integration					
IdP Connections	Require Re-Authentication For Expiring Password Flow	false			
	Require Re-Authentication for Change Password Flow	false			
IdP Adapters	Require Re-Authentication for Password Reset Flow	false			
	Login Challenge Template	html.form.login.challenge.template.html			
Authentication API Applications	'Remember My Username' Lifetime	30			
	'This is My Device' Lifetime	30			
IdP Default URL	Allow Username Edits During Chaining	false			
	Track Authentication Time	true			
	Post-Password Change Re-Authentication Delay	0			
	Password Reset One-Time Link Email Template	message-template-forgot-password-link.html			
	Password Reset One-Time Password Email Template	message-template-forgot-password-code.html			
	Account Disabled Email Template	message-template-account-disabled.html			
	Password Reset Complete Email Template	message-template-forgot-password-complete.html			
	Password Reset Failed Email Template	message-template-forgot-password-failed.html			
	Password Reset Username Template	forgot-password.html			
	Password Reset Code Template	forgot-password-resume.html			
	Password Reset Template	forgot-password-change.html			
	Password Reset Error Template	forgot-password-error.html			
	Password Reset Success Template	forgot-password-success.html			
	Account Unlock Template	account-unlock.html			
	Account Unlock Email Template	message-template-account-unlock-complete.html			
	OTP Length	8			

Figure 25. IdP Adapter summary (part 3): re-authentication, template, and messaging defaults.

Integration	Allowed OTP Character Set	23456789BCDFGHJKLMNPQRSTVWXZbodfghjklmnpqrstvwxyz
IdP Connections	Password Reset Token Validity Time	10
IdP Adapters	PingID Properties	
Authentication API Applications	Require Verified Email	false
IdP Default URL	Username Recovery Template	username.recovery.template.html
	Username Recovery Info Template	username.recovery.info.template.html
	Username Recovery Email Template	message-template-username-recovery.html
	Client Side Authenticator	None Selected
	Risk Provider	None Selected
	Risk for Authentication	false
	Risk for Password Change	false
	Risk for Password Reset	false
	Risk for Username Recovery	false
	Fail Authentication on Account Lockout	false
	Extended Contract	
	Attribute	policy.action
	Attribute	username
	Adapter Attributes	
	Pseudonym	username
	Unique User Key Attribute	username

Figure 26. IdP Adapter summary (part 4): remaining defaults, plus the Extended Contract (*policy.action*, *username*) and Adapter Attributes (*pseudonym* and *unique user key* both set to *username*).

Adapter Contract Mapping	
Attribute Sources & User Lookup	
Data Sources	(None)
Adapter Contract Fulfillment	
policy.action	policy.action (Adapter)
username	username (Adapter)
Issuance Criteria	
Criterion	(None)

Cancel Previous Save

Figure 27. Final Adapter Contract Mapping recap — *policy.action* and *username* both fulfilled by the Adapter — before saving the adapter.

What this does

In this step, we created and configured a dedicated HTML Form IdP Adapter called *Cascade-Portal-Login* to serve as the user-facing authentication component for the *Cascade Internal Portal*. Its purpose is to give PingFederate a mechanism for collecting a user's credentials through an HTML-based login form and then passing those credentials to the configured *CyberCorp-LDAP-Validator*, which validates the submitted username and password against the *PingDirectory* instance configured in Lab 1. This creates the authentication path between the application-facing PingFederate layer and the directory where *CyberCorp* users are stored.

We configured the adapter's *Credential Validator* to use *CyberCorp-LDAP-Validator*, establishing the relationship between the HTML login form and *PingDirectory*. We also configured *username* as the *Unique User Key Attribute*, giving PingFederate a consistent identifier for the authenticated user, and selected *username* as the adapter's *pseudonym* — required because the HTML Form IdP Adapter needs at least one *pseudonym* attribute.

The adapter's contract was intentionally kept simple at this stage: it exposes only *username* and *policy.action*, both fulfilled directly by the adapter, with no additional attribute sources or LDAP lookups. The immediate objective was to establish the basic authentication mechanism first. Attributes such as *cn*, *mail*,

departmentNumber, and title, which already exist in PingDirectory, are retrieved later when the authentication policy maps them in from a second LDAP lookup (see 3.5). Issuance Criteria were also left unconfigured, meaning the adapter does not yet apply any conditional rules — such as allowing or denying authentication based on department, job title, or group membership. Those authorization decisions can be introduced later as part of the Cascade Portal's access-control design.

3.4 Creating the Authentication Policy Contract

The screenshot shows the PingFederate console interface. The left sidebar contains a navigation menu with options: Policies, Fragments, Selectors, Policy Contracts (selected), Sessions, and Local Identity Profiles. The main content area is titled 'Policy Contracts | Authentication Policy Contract'. It has three tabs: 'Contract Info' (active), 'Contract Attributes', and 'Summary'. Below the tabs, a text input field labeled 'CONTRACT NAME' contains the value 'CyberCorp-User-Authen'. At the bottom right, there are 'Cancel' and 'Next' buttons.

Figure 28. Naming the new policy contract *CyberCorp-User-Authentication*.

The screenshot shows the PingFederate console interface, specifically the 'Contract Attributes' tab. The left sidebar is the same as in Figure 28. The main content area is titled 'Policy Contracts | Authentication Policy Contract'. The 'Contract Attributes' tab is active. Below the tabs, a text input field labeled 'Attribute Contract' contains the value 'subject'. Below this, a table titled 'Extend the Contract' lists attributes with their corresponding 'Action' links (Edit | Delete).

Extend the Contract	Action
cn	Edit Delete
departmentNumber	Edit Delete
givenName	Edit Delete
mail	Edit Delete
title	Edit Delete
uid	Edit Delete
username	Edit Delete

At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

Figure 29. Contract Attributes: the default subject attribute, extended with *cn*, *departmentNumber*, *givenName*, *mail*, *title*, *uid*, and *username*.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

< Policies

Policy Contracts | Authentication Policy Contract

Contract Info Contract Attributes Summary

Authentication policy contract summary information.

Authentication Policy Contract

Contract Info

Contract Name CyberCorp-User-Authentication

Contract Attributes

Attribute	subject
Attribute	cn
Attribute	departmentNumber
Attribute	givenName
Attribute	mail
Attribute	title
Attribute	uid
Attribute	username

Cancel Previous Save

Figure 30. Policy contract summary confirming *CyberCorp-User-Authentication* before saving.

What this does

Here we defined *CyberCorp-User-Authentication*, an Authentication Policy Contract. Unlike the IdP Adapter's contract, which is fixed to whatever the adapter itself collects, a policy contract is a reusable, adapter-independent attribute set that an authentication policy can populate from one or more sources and then hand off to any application or SP connection that consumes the policy.

We extended it beyond the default subject attribute to include *cn*, *departmentNumber*, *givenName*, *mail*, *title*, *uid*, and *username* - mirroring the attributes stored on CyberCorp user entries in PingDirectory - so the Cascade Portal, or any future relying application, can receive a consistent, directory-backed identity for the authenticated user.

3.5 Creating the Authentication Policy

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

< Policies

Policies

Policies Default Authentication Sources Tracked HTTP Parameters

Authentication policies define how PingFederate authenticates users. Selectors and authentication sources can be conditionally chained together in paths to form policies. Ensure that successful paths end with Policy Contracts to reuse mapping configuration across protocols and applications.

☐ IDP AUTHENTICATION POLICIES

☐ SP AUTHENTICATION POLICIES

☐ FAIL IF POLICY ENGINE FINDS NO AUTHENTICATION SOURCE

Policy	Authentication Sources	Selectors	Policy Contracts	Fragments	Enabled	Action
PingDirectory-Password-Policy	PingDirectory-HTML		PingDirectory-Username-Contract		☒	Select Action

Add Policy

Cancel Next Save

Figure 31. Policies list, showing the existing *PingDirectory-Password-Policy*, before adding a new policy.

The screenshot shows the PingFederate console interface. The left sidebar contains a navigation menu with options: Policies, Fragments, Selectors, Policy Contracts, Sessions, and Local Identity Profiles. The main content area is titled 'Policies | Policy'. It includes a description of authentication policies and a form to create a new policy. The policy name is 'CyberCorp-Cascade-Authentication', the ID is 'BDZXJJAXbi67KRoz7m6Dowad8', and the description is 'Authentication policy for the Cascade Internal Portal using the CyberCorp HTML Form IdP Adapter and PingDirectory LDAP authentication.' There is a checkbox for 'HANDLE FAILURES LOCALLY' which is unchecked. The 'POLICY' section shows a chain of two rules: 'Cascade-Portal-Login - (Adapter)' followed by 'CyberCorp-User-Authentication - (Polik)'. The 'FAIL' path is set to 'Done' and the 'SUCCESS' path is set to 'CyberCorp-User-Authentication - (Polik)'. At the bottom right, there are 'Cancel' and 'Done' buttons.

Figure 32. New policy *CyberCorp-Cascade-Authentication*, chaining the *Cascade-Portal-Login* adapter to the *CyberCorp-User-Authentication* policy contract on success.

This screenshot shows the same PingFederate console interface as Figure 32, but with an error message at the top: 'Please fix the mappings with errors.' The policy configuration is identical, but the 'SUCCESS' path now shows a red error icon next to the 'CyberCorp-User-Authentication - (Polik)' rule. The 'Contract Mapping' link is also highlighted in red. The 'Cancel' and 'Done' buttons remain at the bottom right.

Figure 33. “Please fix the mappings with errors” — the initial save attempt failed because the adapter alone cannot supply every attribute the policy contract expects.

What this does

The Cascade-Portal-Login adapter's contract only produces username and policy.action - it has no way to supply cn, departmentNumber, givenName, mail, subject, title, or uid on its own. Because the CyberCorp-User-Authentication policy contract expects all of those, PingFederate flagged the Contract Mapping step with an error and refused to save until it was resolved.

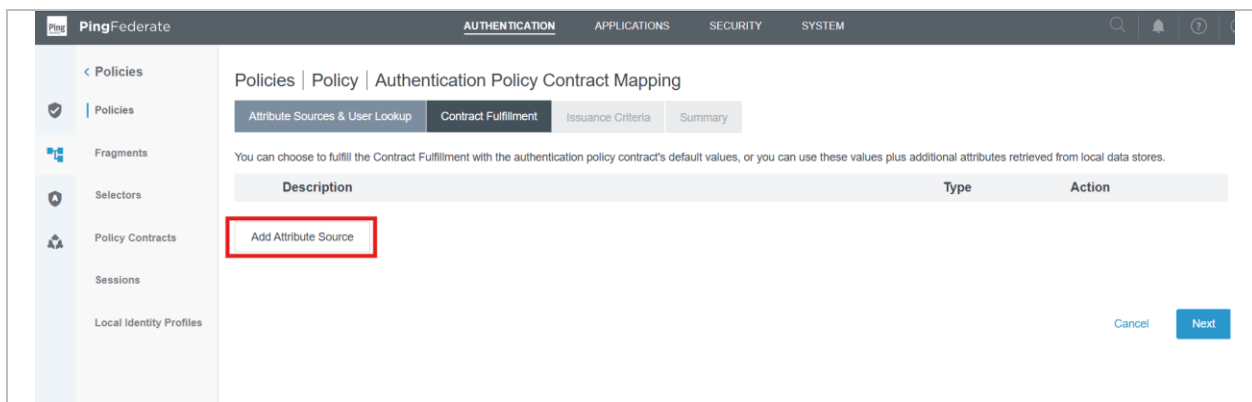


Figure 34. Resolving the mapping: adding a new attribute source to the policy.

Policies | Policy | Authentication Policy Contract Mapping | Attribute Sources & User Lookup

Data Store | LDAP Directory Search | LDAP Filter | Summary

Data stores are used to retrieve supplemental attributes. Specify the attribute store's details to use it in your configuration.

ATTRIBUTE SOURCE ID	CyberCorpPD
ATTRIBUTE SOURCE DESCRIPTION	Retrieve Cascade Portal
ACTIVE DATA STORE	CyberCorp-PD
DATA STORE TYPE	LDAP

Manage Data Stores

Figure 35. Attribute source CyberCorpPD, reusing the same CyberCorp-PD LDAP data store created in Step 3.1.

Policies | Policy | Authentication Policy Contract Mapping | Attribute Sources & User Lookup

Data Store | **LDAP Directory Search** | LDAP Filter | Summary

Please configure your directory search. This information, along with the attributes supplied in the contract, will be used to fulfill the contract.

BASE DN ?

SEARCH SCOPE

Attributes to return from search

ROOT OBJECT CLASS	ATTRIBUTE	OPTION	Action
	departmentNumber		Remove
	givenName		Remove
	mail		Remove
	Subject DN		
	uid		Remove

[View Attribute Contract](#)

[Cancel](#) [Previous](#) [Next](#)

Figure 36. LDAP Directory Search: Base DN `ou=People,dc=cybercorp,dc=local`, Subtree scope, returning `departmentNumber`, `givenName`, `mail`, `Subject DN`, and `uid`.

Policies | Policy | Authentication Policy Contract Mapping | Attribute Sources & User Lookup

Data Store | LDAP Directory Search | **LDAP Filter** | Summary

Please enter a Filter for extracting data from your directory.

FILTER ?

Values

- `${ad.CascadePortalLogin.policy.action}`
- `${ad.CascadePortalLogin.username}`
- `${context.ClientId}`
- `${context.ClientIp}`
- `${context.HttpRequest}`
- `${context.OAuthScopes}`
- `${context.SpConnEntityId}`
- `${context.VirtualServerId}`
- `${context.sri}`

[View List of Available LDAP Attributes](#)

Figure 37. LDAP Filter: `(uid=${ad.CascadePortalLogin.username})` — dynamically re-searching PingDirectory for the user the adapter just authenticated.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

Policies > Policy > Authentication Policy Contract Mapping | Attribute Sources & User Lookup

Data Store LDAP Directory Search LDAP Filter Summary

Attribute Source Summary

Attribute Sources & User Lookup

Data Store

Attribute Source	Retrieve Cascade Portal user attributes from PingDirectory
Attribute Source ID	CyberCorpPD
Type of Data Store	LDAP
Data Store	CyberCorp-PD

LDAP Directory Search

Base DN	ou=People,dc=cybercorp,dc=local
Search scope	SUBTREE_SCOPE
Attribute	departmentNumber
Attribute	givenName
Attribute	mail
Attribute	Subject DN
Attribute	uid

LDAP Filter

Filter	(uid=\${ad.CascadePortalLogin.username})
--------	--

Cancel Previous Done

Figure 38. Attribute source summary confirming the LDAP lookup before returning to the contract mapping.

PingFederate AUTHENTICATION APPLICATIONS SECURITY SYSTEM

Policies > Policy > Authentication Policy Contract Mapping

Attribute Sources & User Lookup Contract Fulfillment Issuance Criteria Summary

Fulfill your Authentication Policy Contract with values from the authentication sources or with dynamic text values.

Adapter Contract	Source	Value	Actions
cn	No Mapping		None available
departmentNumber	LDAP (Retrieve Cascade Portal user attributes from PingDirectory)	departmentNumber	None available
givenName	LDAP (Retrieve Cascade Portal user attributes from PingDirectory)	givenName	None available
mail	LDAP (Retrieve Cascade Portal user attributes from PingDirectory)	mail	None available
subject	LDAP (Retrieve Cascade Portal user attributes from PingDirectory)	Subject DN	None available
title	No Mapping		None available
uid	LDAP (Retrieve Cascade Portal user attributes from PingDirectory)	uid	None available
username	Adapter (CascadePortalLogin)	username	None available

Cancel Previous Next

Figure 39. Contract Fulfillment: departmentNumber, givenName, mail, subject, and uid mapped from the new LDAP lookup; username mapped from the Adapter; cn and title left as No Mapping.

Note: cn and title are left unmapped because the LDAP Directory Search in this step wasn't configured to retrieve them - only departmentNumber, givenName, mail, Subject DN, and uid were added. This is a known gap: either add cn and title to the search's returned attributes, or remove them from the policy contract, before treating this contract as complete.

Policies | Policy | Authentication Policy Contract Mapping

Attribute Sources & User Lookup Contract Fulfillment Issuance Criteria **Summary**

Summary of Authentication Policy Contract Mapping

Authentication Policy Contract Mapping

Attribute Sources & User Lookup

Data Store	Retrieve Cascade Portal user attributes from PingDirectory (LDAP)
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Attribute Sources & User Lookup

Data Store	
Attribute Source	Retrieve Cascade Portal user attributes from PingDirectory
Attribute Source ID	CyberCorpPD
Type of Data Store	LDAP
Data Store	CyberCorp-PD

LDAP Directory Search

Base DN	ou=People,dc=cybercorp,dc=local
Search scope	SUBTREE_SCOPE
Attribute	departmentNumber
Attribute	givenName
Attribute	mail
Attribute	Subject DN
Attribute	uid

LDAP Filter

Filter	(uid=\${ad.CascadePortalLogin.username})
--------	--

Contract Fulfillment

cn	No Mapping
departmentNumber	departmentNumber (LDAP)
givenName	givenName (LDAP)
mail	mail (LDAP)
subject	Subject DN (LDAP)
title	No Mapping
uid	uid (LDAP)
username	username (Adapter)

Issuance Criteria

Criterion	(None)
-----------	--------

[Cancel](#) [Previous](#) [Done](#)

Figure 40. Full Authentication Policy Contract Mapping summary, showing the attribute source, LDAP search, filter, and contract fulfillment together.

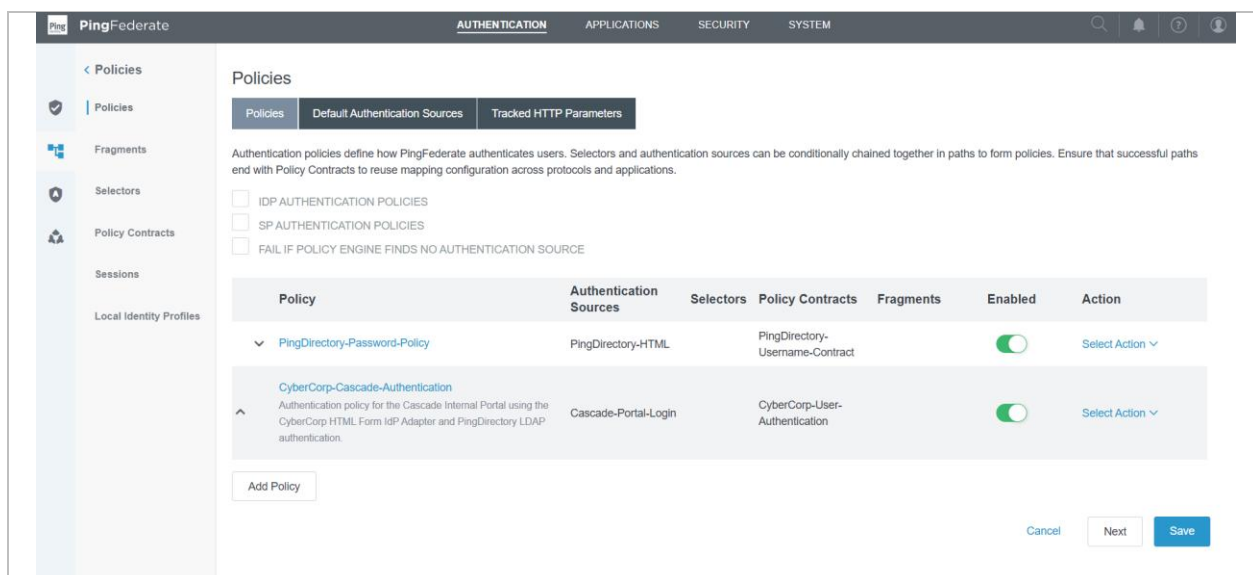


Figure 41. The Policies list after saving — *CyberCorp-Cascade-Authentication* now appears alongside *PingDirectory-Password-Policy*, enabled.

What this does

The Authentication Policy is what actually ties everything together into a login flow: it chains *Cascade-Portal-Login* (which collects and validates the user's credentials via *CyberCorp-LDAP-Validator*) to *CyberCorp-User-Authentication* (which carries the user's attributes forward to any relying application).

We resolved the initial mapping error by adding a second attribute source to the policy — a fresh LDAP lookup against the same *CyberCorp-PD* data store, searching `ou=People,dc=cybercorp,dc=local` with the filter `(uid=${ad.CascadePortalLogin.username})` to re-fetch the just-authenticated user's record and pull `departmentNumber`, `givenName`, `mail`, `Subject DN`, and `uid` directly from *PingDirectory*.

With `departmentNumber`, `givenName`, `mail`, `subject`, and `uid` mapped from that LDAP lookup and `username` mapped from the adapter, the policy saved successfully and now appears enabled alongside the lab's original *PingDirectory-Password-Policy*. `cn` and `title` remain an open item, tracked in the note above.

4. Summary

By the end of this lab, *PingFederate* is wired to the *CyberCorp PingDirectory* environment from Lab 1. A *CyberCorp-PD* LDAP data store and a *CyberCorp-LDAP-Validator* credential validator give *PingFederate* the ability to check submitted credentials against `ou=People,dc=cybercorp,dc=local` using the `svc-pingfederate` service account. A *Cascade-Portal-Login* HTML Form IdP Adapter presents the login form and authenticates the user through that validator, exposing `username` and `policy.action`.

A *CyberCorp-User-Authentication* policy contract defines the full set of user attributes the *Cascade Portal* expects, and the *CyberCorp-Cascade-Authentication* policy chains the adapter to that contract, using a second LDAP lookup to fill in `departmentNumber`, `givenName`, `mail`, `subject`, and `uid` from *PingDirectory*. `cn` and `title` remain unmapped and are flagged as a follow-up item. With the policy saved and enabled, *PingFederate* is ready to authenticate *CyberCorp* users for the *Cascade Internal Portal*.